

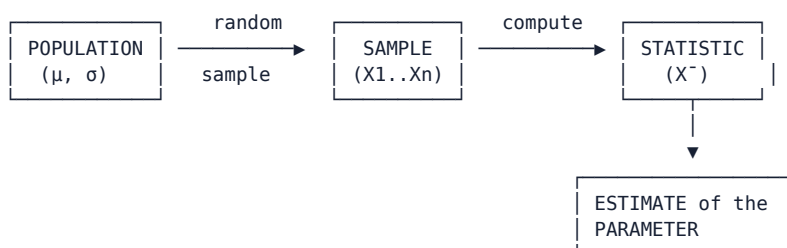
09 THEORY OF ESTIMATION (ppt5, 26 pages, lectures 19-21)

1. The vocabulary (p003-p004)

PARAMETER	a numerical characteristic of a POPULATION μ, σ^2, p
STATISTIC	a number computed FROM SAMPLE DATA $\bar{X}, S^2, \hat{p}$
ESTIMATOR	a FUNCTION of the sample used to estimate a parameter. It is a RANDOM VARIABLE. e.g. $\bar{X}$ as an estimator of μ
ESTIMATE	the NUMBER you get once data is plugged in e.g. $\bar{x} = 52$
PARAMETER SPACE	all possible values of the parameter $\theta = (-\infty, \infty)$ for μ

ESTIMATOR vs ESTIMATE - one mark, easy to lose
 $\bar{X}$ = the RULE (an estimator, a random variable)
 $\bar{x} = 205$ = the NUMBER (the estimate, once data is in)

2. The estimation pipeline (p004)



3. Point vs interval estimation (p005-p007)

POINT ESTIMATION

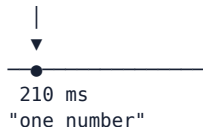
a SINGLE value

$$\mu \approx \bar{X} \quad \sigma^2 \approx S^2 \quad p \approx \hat{p}$$

- ✓ simple, easy to report
- ✗ does not show uncertainty

the picture from p007:

POINT



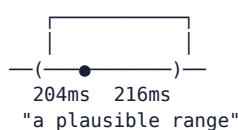
INTERVAL ESTIMATION

a RANGE of values

with a stated confidence level

- ✓ accounts for uncertainty
- ✓ more informative

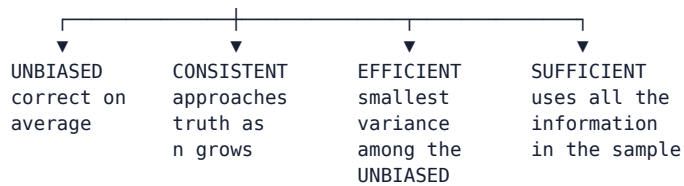
INTERVAL



$$CI = \text{Point Estimate} \pm \text{Margin of Error}$$

4. The four characteristics of a good estimator (p008)

A GOOD ESTIMATOR

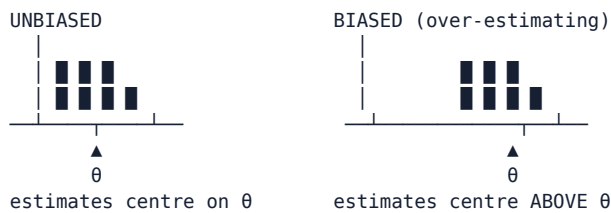


"good = little systematic error + stability + precision"

5. Unbiasedness (p009)

$\hat{\theta}$ is UNBIASED for θ if $E(\hat{\theta}) = \theta$
 $\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$

the picture: many samples, plot the estimates



"CORRECT ON AVERAGE, NOT PER SAMPLE"
 An unbiased estimator can be badly wrong on any ONE sample.
 Unbiasedness is a statement about the LONG RUN.

6. Consistency (p010-p012)

$\hat{\theta}_n$ converges in probability to θ as $n \rightarrow \infty$
 $\hat{\theta}_n \xrightarrow{P} \theta$

i.e. for any $\epsilon > 0$: $P(|\hat{\theta}_n - \theta| < \epsilon) \rightarrow 1$ as $n \rightarrow \infty$

SUFFICIENT CONDITIONS FOR CONSISTENCY (a sequence T_n):

1. $E(T_n) \rightarrow \theta$ as $n \rightarrow \infty$
2. $\text{Var}(T_n) \rightarrow 0$ as $n \rightarrow \infty$

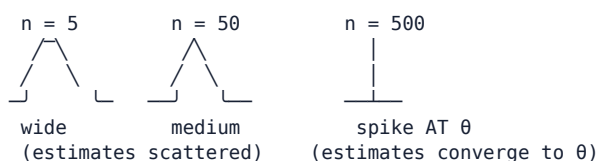
Both hold $\Rightarrow T_n$ is consistent.

WHY $\bar{X}$ IS CONSISTENT (p012) - the standard answer:

CONDITION 1	CONDITION 2	CONCLUSION
$E(\bar{X}) = \mu$ so $E(\bar{X}) \rightarrow \mu$	$\text{Var}(\bar{X}) = \sigma^2/n$ so $\sigma^2/n \rightarrow 0$	both hold $\Rightarrow \bar{X}$ is consistent

"more observations $\rightarrow$ smaller variance $\rightarrow$ more stable estimate"

the picture:



7. Efficiency (p013-p014)

$\hat{\theta}^1$ and $\hat{\theta}^2$ both UNBIASED for θ :
 if $\text{Var}(\hat{\theta}^1) < \text{Var}(\hat{\theta}^2)$ then $\hat{\theta}^1$ is MORE EFFICIENT

Among all UNBIASED estimators, the most efficient has the SMALLEST variance.
 Efficiency $\propto 1/\text{Variance}$

the slide's visual:

Estimator A
unbiased, Var = 4



concentrated
higher precision
-> MORE EFFICIENT

Estimator B
unbiased, Var = 9



spread out
lower precision
-> less efficient

THE ORDER OF OPERATIONS IN ANY COMPARISON:

- STEP 1 find which are UNBIASED (compute E of each)
- STEP 2 a biased one is ELIMINATED, even if its variance is smaller ("efficiency" only compares the unbiased)
- STEP 3 compare variances of the SURVIVORS, smallest wins

8. WORKED COMPARISON SET 1 (p015-p018) - the flagship question

Sample of size 5 from $N(\mu, \sigma^2)$. ($X_1..X_5$)

```
t1 = (X1+X2+X3+X4+X5)/5      <- the sample mean
t2 = (X1+X2)/2 + X3
t3 = (2X1 + X2 + λX3)/3      <- λ chosen for unbiasedness
```

STEP 1: FIND λ

$E(t3) = [2E(X1) + E(X2) + \lambda E(X3)]/3$
 $= (2\mu + \mu + \lambda\mu)/3 = [(3+\lambda)/3] \mu$
 for unbiasedness: $(3+\lambda)/3 = 1 \rightarrow \lambda = 0$
 so $t3 = (2X1 + X2)/3$

STEP 2: WHICH ARE UNBIASED

$E(t1) = (\mu+\mu+\mu+\mu+\mu)/5 = \mu$ ✓ unbiased
 $E(t2) = (\mu+\mu)/2 + \mu = 2\mu$ ✗ BIASED
 $E(t3) = (2\mu+\mu)/3 = \mu$ ✓ unbiased

STEP 3: COMPARE THE VARIANCES (of the unbiased ones)

$\text{Var}(t1) = \sigma^2/5 = 0.2000 \sigma^2$
 $\text{Var}(t3) = (4\sigma^2 + \sigma^2)/9 = 5\sigma^2/9 = 0.5556 \sigma^2$

ordering: $\sigma^2/5 < 5\sigma^2/9 < 3\sigma^2/2$
 (t1) (t3) (t2, biased)

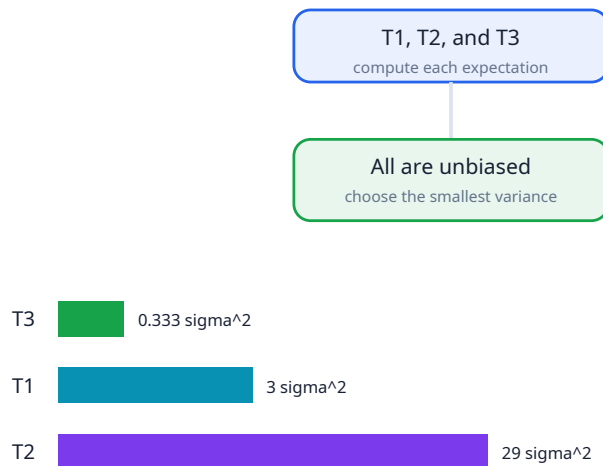
BEST = $t1 = \bar{X}$, the sample mean

the variance bar chart:

t1		0.200 σ^2	<- WINNER
t3		0.556 σ^2	
t2		1.500 σ^2	(biased, disqualified)

9. WORKED COMPARISON SET 2 (p019-p021) - with consistency

Estimator set 2: all unbiased, compare variance



Consistency needs a sequence T_n as n grows; fixed $n=3$ cannot prove it.

Unbiased estimators compared by variance

Sample of size 3. T_1, T_2, T_3 estimate μ .

$$T_1 = X_1 + X_2 - X_3$$

$$T_2 = 2X_1 + 3X_3 - 4X_2$$

$$T_3 = (\lambda X_1 + X_2 + X_3)/3$$

(i) ARE T_1 and T_2 UNBIASED?

$$E(T_1) = \mu + \mu - \mu = \mu \quad \checkmark$$

$$E(T_2) = 2\mu + 3\mu - 4\mu = \mu \quad \checkmark$$

BOTH unbiased

(ii) FIND λ FOR T_3

$$E(T_3) = (\lambda\mu + \mu + \mu)/3 = ((\lambda+2)/3)\mu$$

$$((\lambda+2)/3)\mu = \mu \rightarrow \lambda = 1$$

so $T_3 = (X_1 + X_2 + X_3)/3 = \bar{X}$, the sample mean

(iii) IS T_3 CONSISTENT?

With $\lambda=1$, this fixed- n statistic is the sample mean for $n=3$.

That fact alone does not establish consistency. The sequence $\bar{X}_n$ is consistent under the usual iid assumptions because $\text{Var}(\bar{X}_n) = \sigma^2/n \rightarrow 0$.

(iv) VARIANCES

$$\text{Var}(T_1) = \text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3) = 3\sigma^2$$

$$\text{Var}(T_2) = 4\text{Var}(X_1) + 9\text{Var}(X_3) + 16\text{Var}(X_2) = 29\sigma^2$$

$$\text{Var}(T_3) = (1/9)(3\sigma^2) = \sigma^2/3$$

minimum $\rightarrow T_3$ is the BEST estimator

the variance bar chart:



THE CONSISTENCY TRAP (worth memorising):
 T_1 's variance is $3\sigma^2$ for a FIXED $n=3$.
 That is NOT a consistency statement.
 Consistency is about behaviour AS n GROWS.
 A fixed- n variance, however small or large, says nothing about consistency.

10. Sufficiency (the lms-theory extra)

an estimator is SUFFICIENT if it uses ALL the information about θ present in the sample
the test is the NEYMAN-FISHER factorisation

examples in the corpus: the Poisson sample sum, the exponential sample mean, the sufficient statistics MCQ on MTE 2025-26 Q3

11. WORKED Real-Life 1: response time (p019)

response times of 8 requests: 180,210,195,220,205,190,200,240

$$\bar{x} = (\text{sum})/8 = 1640/8 = 205 \text{ ms}$$

ESTIMATOR = $\bar{x}$ (the rule)
ESTIMATE = 205 ms (the number)

THE QUESTION ASKS FOR **BOTH** "the estimator and the estimate". Two marks. Do not give only the number.

12. WORKED Real-Life 2: packet success (p020)

500 packets, 465 delivered. Estimate p .

$$\hat{p} = X/n = 465/500 = 0.93 = 93\%$$

estimator used: $\hat{p}$

13. WORKED Real-Life 3: battery life (p021) - the CI boundary item

$n = 25$, $\bar{x} = 8.4$ hours, $\sigma = 1.5$ known. 95% CI, $z(0.975) = 1.96$

This z interval assumes a normal population or a separately justified normal approximation. A sample size of 25 alone does not supply that justification.

$$\begin{aligned} \text{CI} &= \bar{x} \pm z(\sigma/\sqrt{n}) \\ &= 8.4 \pm 1.96(1.5/5) \\ &= 8.4 \pm 1.96(0.3) \\ &= 8.4 \pm 0.588 \\ &= (7.81, 8.99) \text{ hours} \end{aligned}$$

the picture:



NOTE: the CI MECHANICS belong to lecture 25 (BOUNDARY for this audit), but "point and interval estimation" IS named in the MTE syllabus line 15. Keep the formula in reserve; do not drill the mechanics.

14. The comparison quick table (p022)

#	PARAMETER	STATISTIC	RESULT
1	μ	$\bar{x}$	205 ms
2	p	$\hat{p}$	0.93
3	μ	$\bar{x}$	95% CI (7.81,8.99)

point -> one value interval -> a range

15. The standard unbiased results (the toolkit)

ESTIMATOR	ESTIMATES	VARIANCE
$\bar{X}$	μ	σ^2/n
$S^2 = \sum (X_i - \bar{X})^2 / (n-1)$	σ^2	(the n-1 matters)
$\hat{p} = X/n$	p	$p(1-p)/n$

WHY n-1 AND NOT n IN S^2

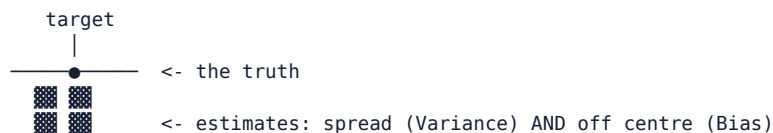
The deviations are measured from $\bar{X}$, which ALREADY used the data. Using n would make S^2 too small on average. Dividing by n-1 corrects that. It is the price of estimating the mean from the same data.

16. MSE, the honest comparison (formula sheet addition)

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

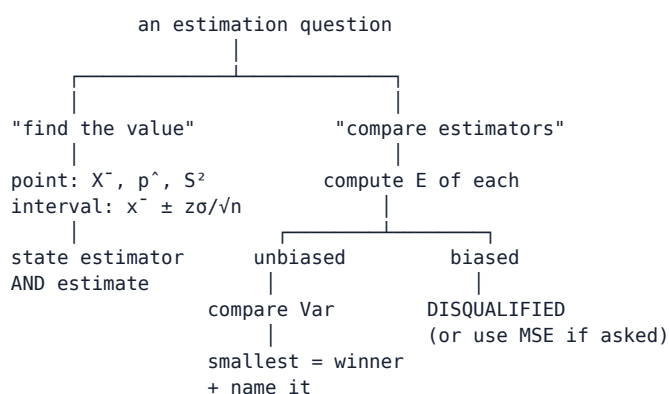
the picture:

$$\text{MSE} = \underbrace{\text{VARIANCE}}_{\text{spread}} + \underbrace{\text{BIAS}^2}_{\text{off-target}}$$



Efficiency ONLY compares unbiased estimators.
MSE lets a biased estimator compete, by paying for its bias in the second term. The exam could introduce this.

THE ESTIMATION DECISION TREE (for any question in this block)



That is the theory of estimation. The estimator-comparison layout (unbiased -> variances -> name the winner) is the 6-to-8 mark question of this block.